

Computational Fluid Dynamics of a Buoyancy-Driven Chimney System

Overview:

The project modelled buoyancy-driven flow through a two-dimensional heat chamber-chimney system. Hot air rises through the channel and chimney while cooler air is drawn in through lateral inlets, a natural-convection configuration modelled with the Boussinesq approximation. The chamber-chimney mechanism is shown below:

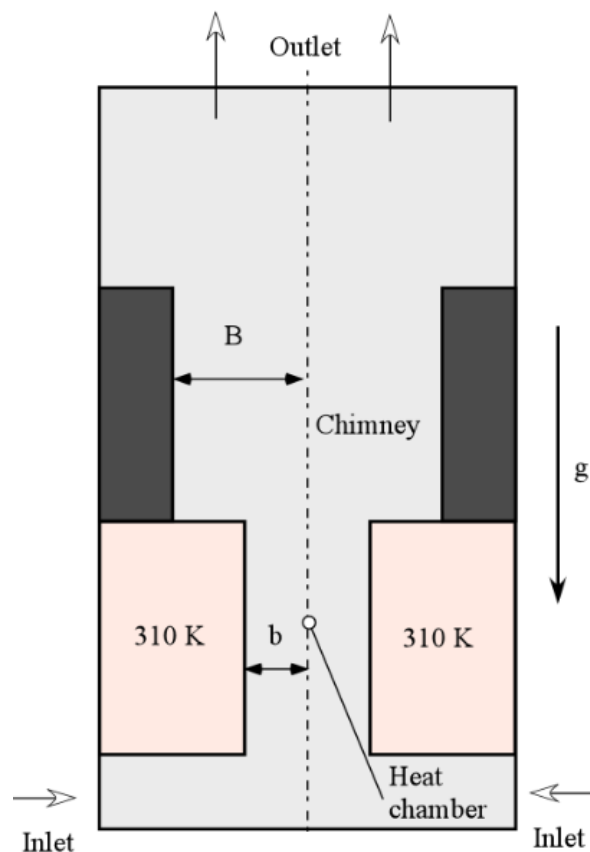


Figure 1: Heat Chamber-Chimney Mechanism

The aim was to build and validate the CFD model, establish a grid-independent mesh and critically compare the pressure-based and density-based solvers and the turbulence-modelling options.

A two-dimensional geometry model of the heat chamber-chimney was created in ANSYS DesignModeler and simulated in ANSYS Fluent using a steady-state analysis followed by a transient analysis. A total of nine linked studies were conducted. The Reynolds, Rayleigh and Mach numbers were hand calculated throughout to verify the modelling assumptions.

Key Results:

The face meshing feature was used to discretise the domain with quadrilateral elements, giving an orthogonal quality of 1.0 and negligible skewness, confirming numerical stability. The chosen mesh from the grid independence study is shown below:

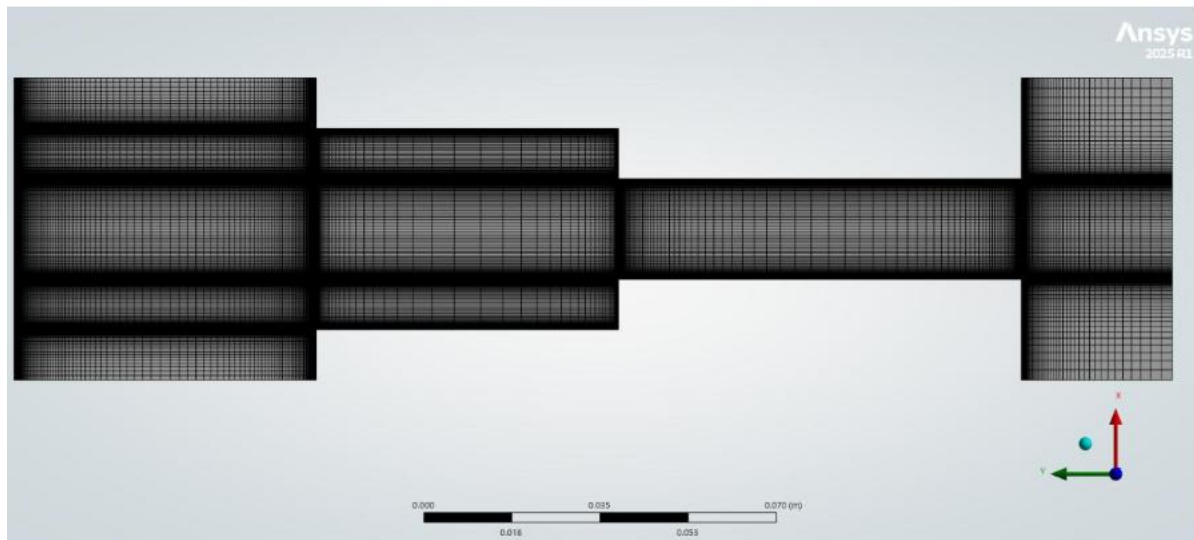


Figure 2: Longitudinal View of chosen mesh from Grid Independence Study

The model captured a central buoyancy-driven high-velocity jet rising through the channel and chimney, with a peak velocity of 0.496m/s. This is shown in **Figure 3**.

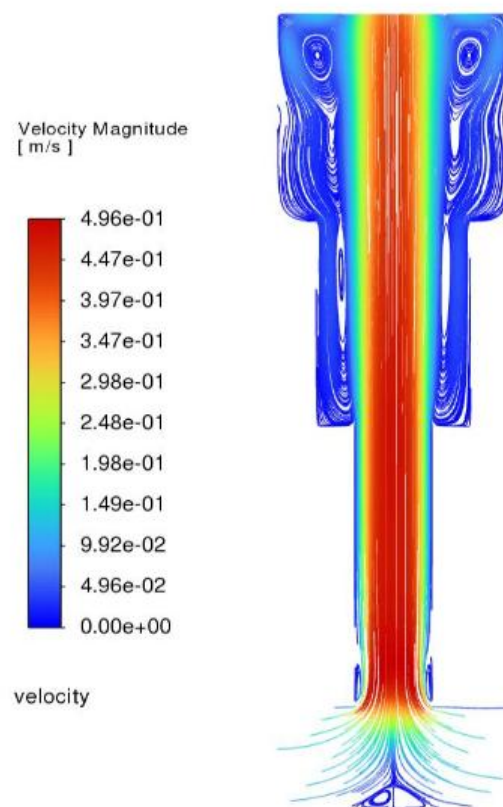


Figure 3: Velocity Magnitude Pathlines

The solver assumptions were verified against computed Rayleigh, Reynolds and Mach numbers.

The pressure-based and density-based solvers produced near-identical velocity magnitudes; however, the density-based solver required ~88% more computation time. The pressure-based solver converged more smoothly and produced a more developed velocity pathline, as illustrated in **Figures 4 & 5**.

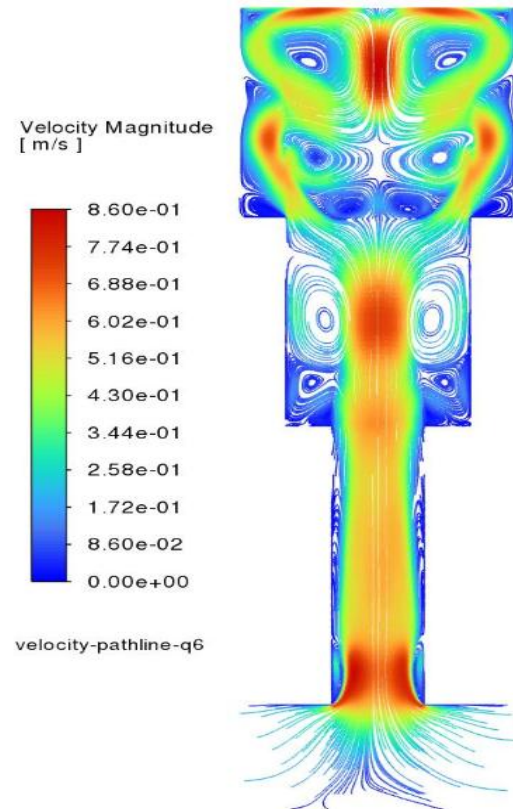
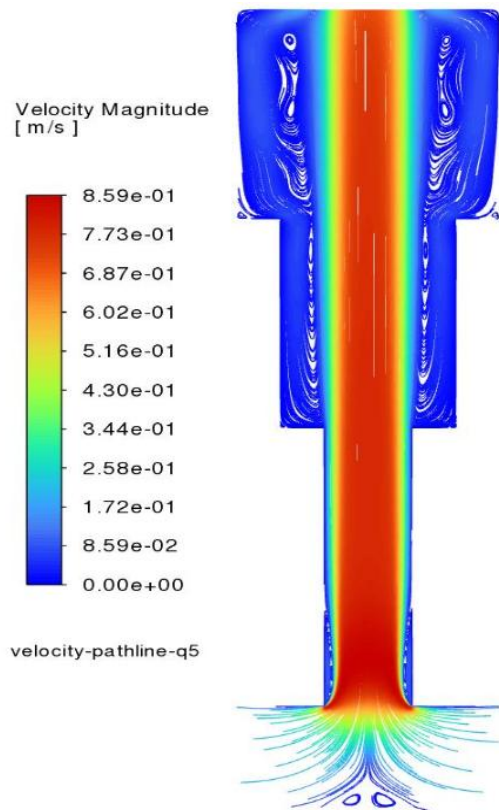


Figure 4: Velocity Magnitude for Pressure-Based Solver

Figure 5: Velocity Magnitude for Density-Based Solver

Note: The boundary conditions were changed when conducting the pressure-based vs density-based solver study, thus explaining the difference between the velocity magnitudes found in **Figure 3** and **Figures 4 & 5**.

The flow operates within a highly turbulent regime, characterised by a pulsating central jet and the shedding of multiple eddies. A DNS was shown to be impractical, so a Large Eddy Simulation with the Smagorinsky-Lilly sub-grid model was recommended over RANS models.